Lab 6:

Code and Branch Coverage Analysis

L6_L7_37_irrigation_scheduler

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Irrigation Scheduler – Testing Report

1. Aim

To test the Irrigation Scheduler Python application using Pytest and measure code coverage and branch coverage using Coverage.py.

2. Procedure

1. **Created test cases for the irrigation scheduler.**
2. **Executed the test cases using Pytest.**
3. **Measured code coverage using Coverage.py.**
4. **Enabled branch coverage and generated the final report.**

5. **Checked the test results and coverage percentages.**

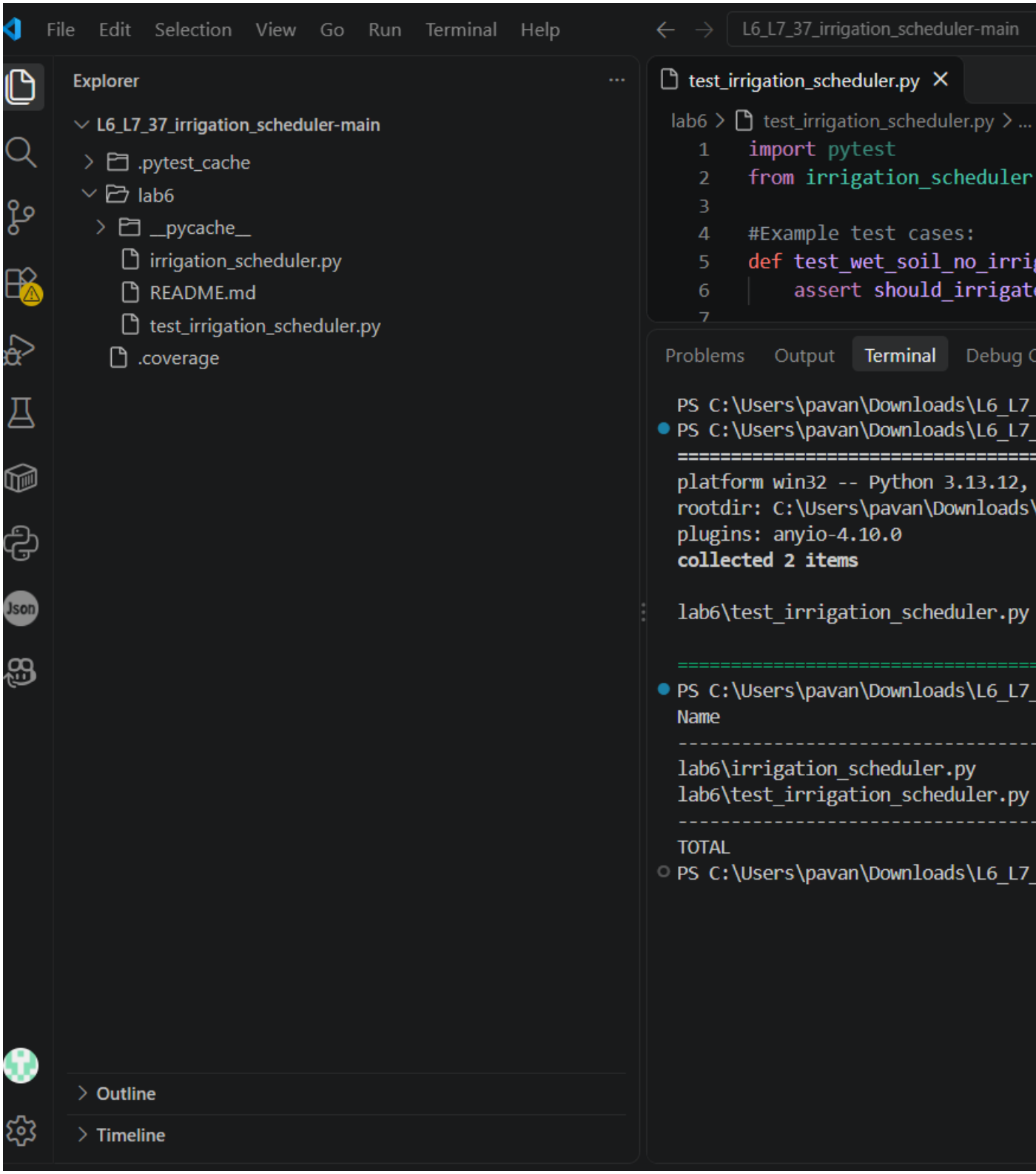
3. Results

- **Total test cases: 18**
- **Tests passed: 18**
- **Tests failed: 0**
- **Statement coverage: 100%**
- **Total statements: 80**
- **Missed statements: 0**
- **Branches reported: 28**
- **Partial branches: 0**
- **Total coverage in the branch-enabled report: 100%**

4. Conclusion

All 18 test cases passed successfully. The coverage report showed 100% statement coverage and 100% total coverage in the branch-enabled report. This indicates that the measured statements and branch outcomes were covered by the tests.

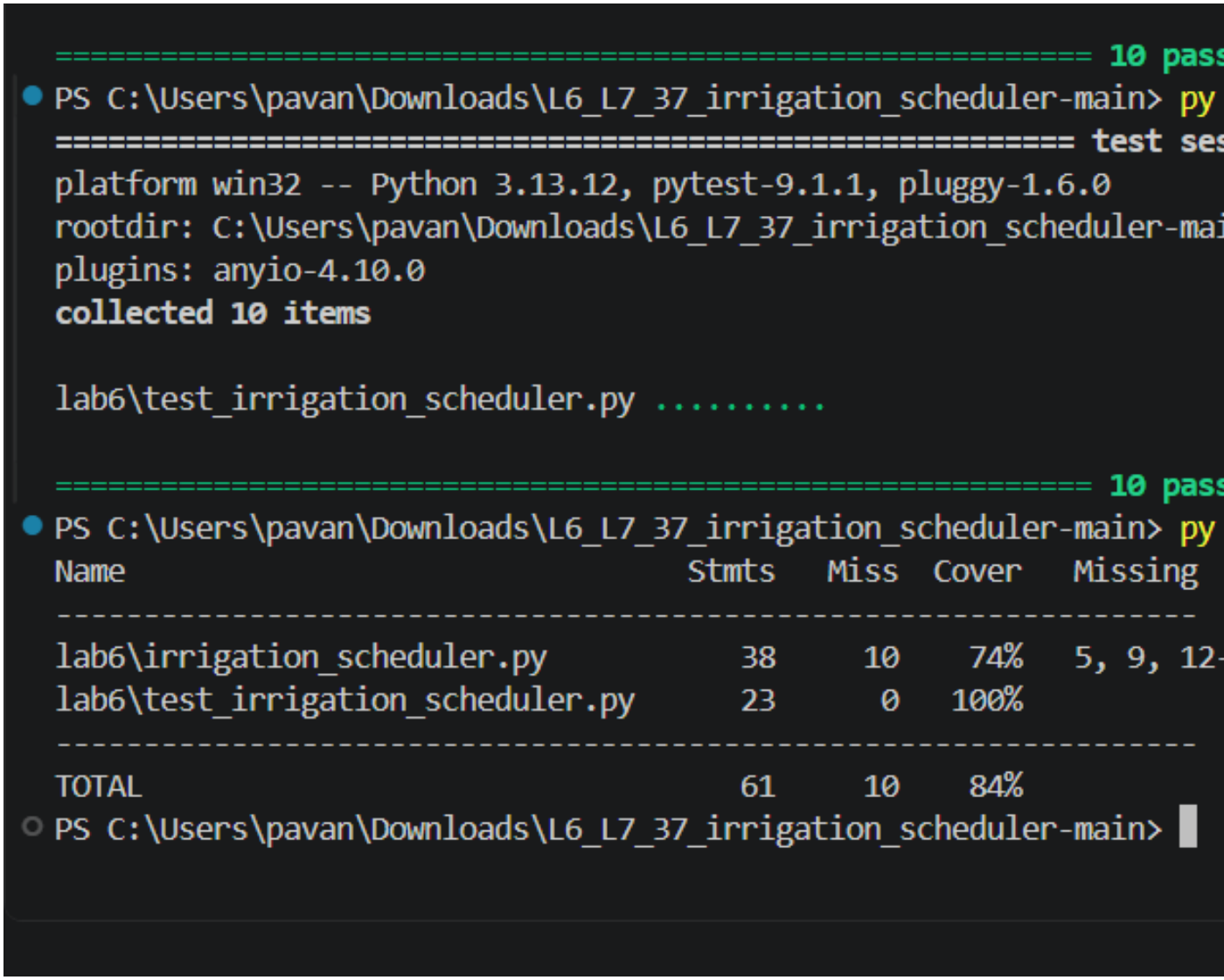
1.


File Edit Selection View Go Run Terminal Help
L6_L7_37_irrigation_scheduler-main
Explorer
L6_L7_37_irrigation_scheduler-main
.pytest_cache
lab6
__pycache__
irrigation_scheduler.py
README.md
test_irrigation_scheduler.py
.coverage
Outline
Timeline
test_irrigation_scheduler.py
lab6 > test_irrigation_scheduler.py > ...
1 import pytest
2 from irrigation_scheduler
3
4 #Example test cases:
5 def test_wet_soil_no_irri
6 assert should_irrigat
Problems Output Terminal Debug
PS C:\Users\pavan\Downloads\L6_L7_
PS C:\Users\pavan\Downloads\L6_L7_
platform win32 -- Python 3.13.12,
rootdir: C:\Users\pavan\Downloads\
plugins: anyio-4.10.0
collected 2 items
lab6\test_irrigation_scheduler.py
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lab6\test_irrigation_scheduler.py
TOTAL
PS C:\Users\pavan\Downloads\L6_L7_

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PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
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platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0 -- C:
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rootdir: C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-ma
plugins: anyio-4.10.0
collected 10 items

lab6/test_irrigation_scheduler.py::test_wet_soil_no_irrigation PAS
lab6/test_irrigation_scheduler.py::test_seedling_volume PASSED
lab6/test_irrigation_scheduler.py::test_seedling_needs_water PASSE
lab6/test_irrigation_scheduler.py::test_flowering_boundary PASSED
lab6/test_irrigation_scheduler.py::test_high_temperature PASSED
lab6/test_irrigation_scheduler.py::test_low_temperature PASSED
lab6/test_irrigation_scheduler.py::test_seedling_normal_temperatur
lab6/test_irrigation_scheduler.py::test_flowering_normal_temperatu
lab6/test_irrigation_scheduler.py::test_schedule_empty_zones PASSE
lab6/test_irrigation_scheduler.py::test_schedule_zero_budget PASSE

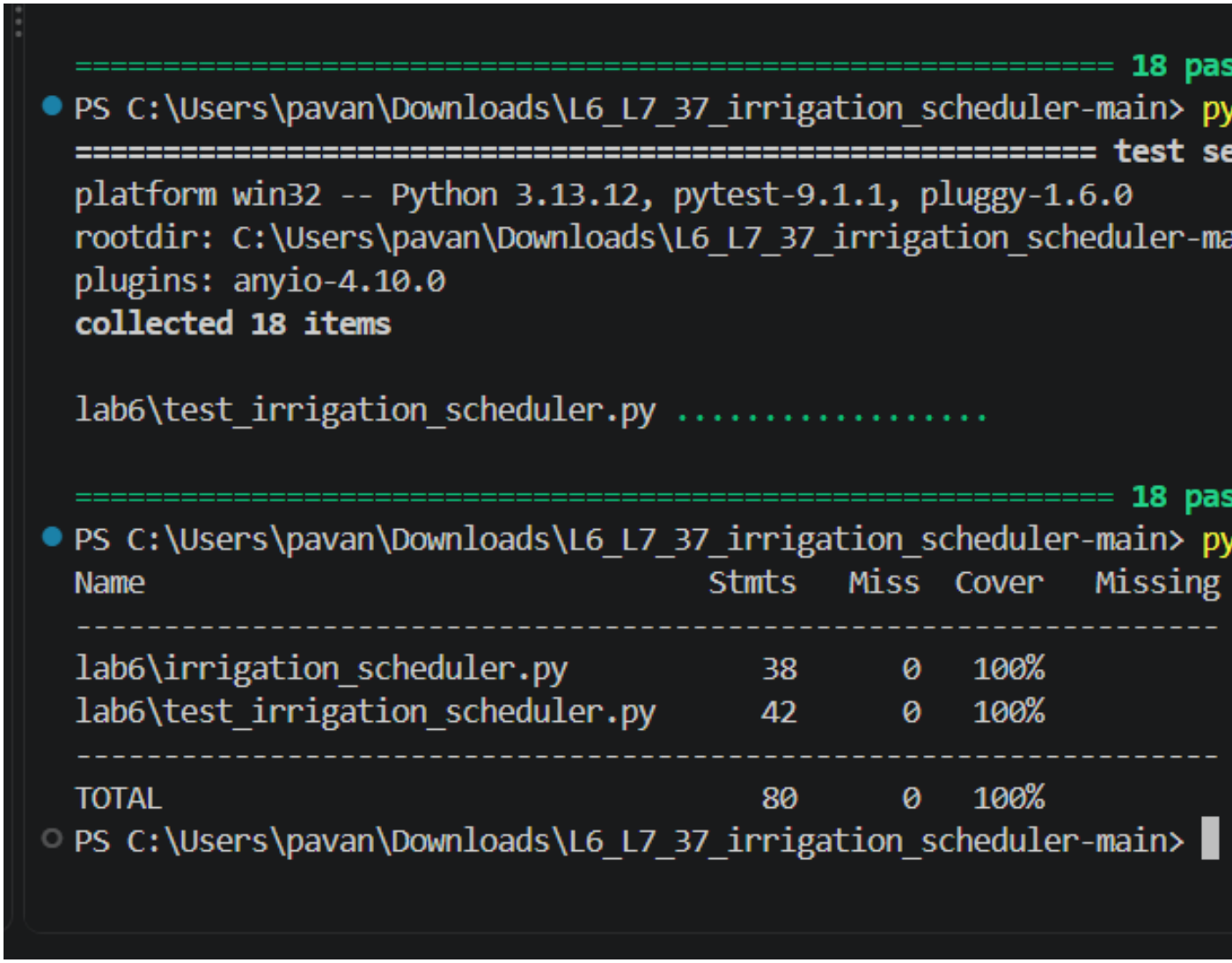
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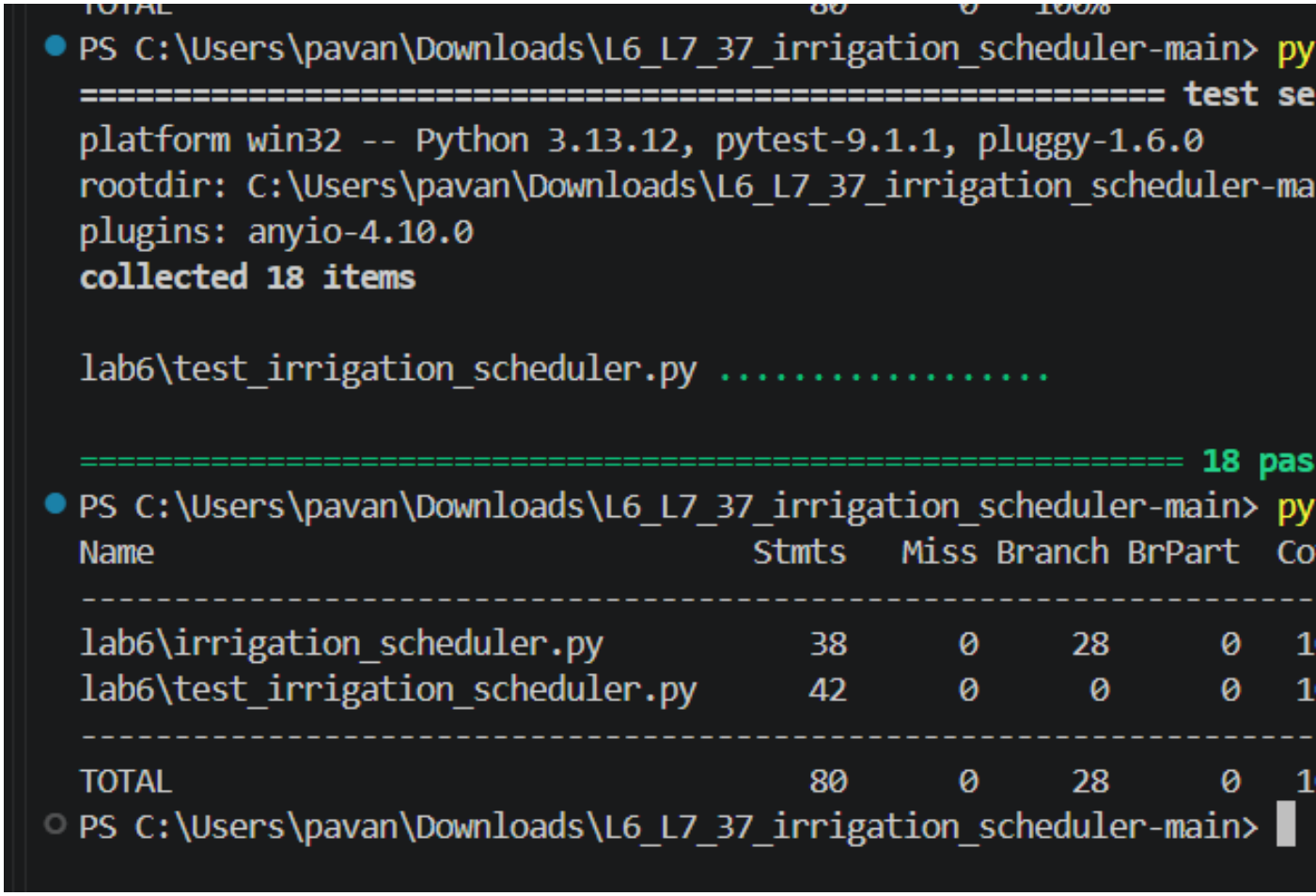

========== 10 pass
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
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platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0
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plugins: anyio-4.10.0
collected 10 items
lab6\test_irrigation_scheduler.py ....
========== 10 pass
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
Name Stmts Miss Cover Missing
lab6\irrigation_scheduler.py 38 10 74% 5, 9, 12-
lab6\test_irrigation_scheduler.py 23 0 100%
TOTAL 61 10 84%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>

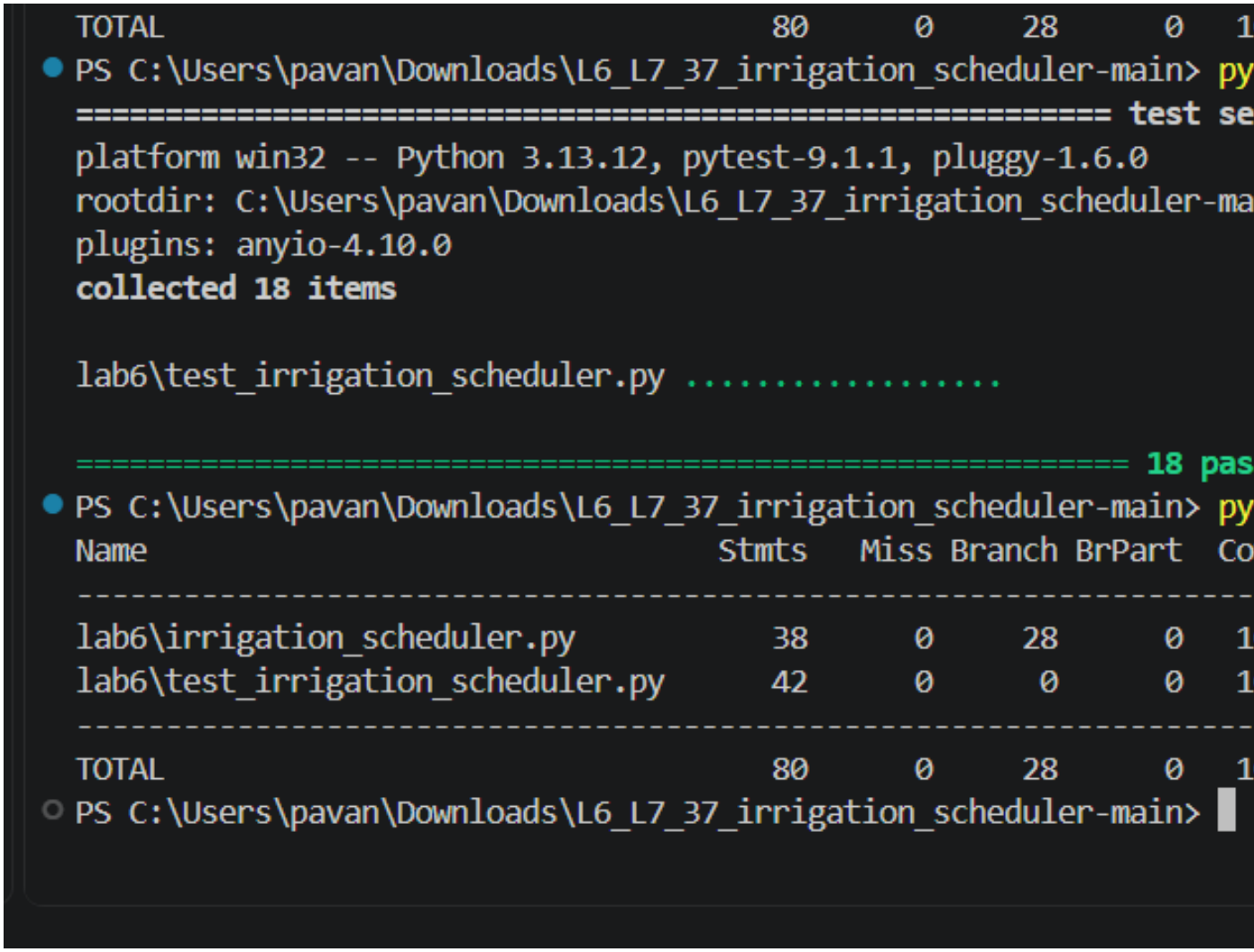
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PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
Name                                  Stmts   Miss  Cover   Missing
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lab6\irrigation_scheduler.py             38     10    74%   5, 9, 12
lab6\test_irrigation_scheduler.py        23      0   100%
-------------------------------------------------------------------
TOTAL                                    61     10    84%
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plugins: anyio-4.10.0
collected 10 items

lab6\test_irrigation_scheduler.py ..........

============================================================ 10 pas
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
Name                                  Stmts   Miss Branch BrPart  Co
-------------------------------------------------------------------
lab6\irrigation_scheduler.py             38     10     28      7
lab6\test_irrigation_scheduler.py        23      0      0      0   1
-------------------------------------------------------------------
TOTAL                                    61     10     28      7
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>
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========== 18 pas
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
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platform win32 -- Python 3.13.12, pytest-9.1.1, pluggy-1.6.0
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lab6\test_irrigation_scheduler.py ....
========== 18 pas
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Name Stmts Miss Cover Missing
lab6\irrigation_scheduler.py 38 0 100%
lab6\test_irrigation_scheduler.py 42 0 100%
TOTAL 80 0 100%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>


TOTAL 80 0 100%
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
test se
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collected 18 items
lab6\test_irrigation_scheduler.py ....
18 pas
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main> py
Name Stmts Miss Branch BrPart Co
lab6\irrigation_scheduler.py 38 0 28 0 1
lab6\test_irrigation_scheduler.py 42 0 0 0 1
TOTAL 80 0 28 0 1
PS C:\Users\pavan\Downloads\L6_L7_37_irrigation_scheduler-main>


TOTAL 80 0 28 0 1
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lab6\test_irrigation_scheduler.py 42 0 0 0 1
TOTAL 80 0 28 0 1
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